Pharmacy Cleanroom Protocol

Step-by-step procedures for sterile compounding (aligned with USP <797>, 2023 revision)

Facility	SOP No.	Effective date	Approved by

How to use this document. This is a template protocol. USP <797> requires each facility to define its own garbing order, cleaning agents and frequencies in written SOPs based on its room layout (e.g., sink location) and the CSP categories it prepares. Review against the official USP <797> text, USP <800> (if hazardous drugs are handled), your state board of pharmacy rules, and manufacturer instructions for your PECs and disinfectants before adopting.

1. Scope and Room Layout

Applies to all personnel entering the anteroom, buffer room (ISO Class 7) or working in a primary engineering control (PEC, ISO Class 5) such as a laminar airflow workbench or biological safety cabinet. Movement is always from **dirty to clean**:

Zone	Air quality	What happens here
General pharmacy	Unclassified	Remove personal items; stage supplies; remove shipping cartons.
Anteroom	ISO Class 8 (ISO 7 if it opens to a negative-pressure room)	Hand hygiene, donning garb (dirty side to clean side line), wiping supplies.
Buffer room	ISO Class 7	Houses the PEC; alcohol hand rub and sterile gloves; staging.
PEC (hood / BSC)	ISO Class 5	Aseptic manipulation of CSPs only.

2. Before Entering the Compounding Area

1	Do not enter if you have a rash, sunburn, weeping sore, conjunctivitis, active respiratory infection or other condition that could contaminate CSPs. Report it to your supervisor.
2	Remove outer garments: coats, hats, jackets, sweaters, vests, scarves.
3	Remove all hand, wrist and other visible jewelry and piercings that could interfere with garb (rings, watches, bracelets, earrings, lanyards).
4	Remove cosmetics (makeup, perfume, hair spray) — they shed particles.
5	Artificial nails, extenders and nail polish are not permitted. Keep natural nails neat and trimmed.
6	Leave phones, earbuds, food, drinks, gum and candy outside. Nothing is eaten, drunk or stored in the compounding area.
7	Remove ear buds/headphones and wipe eyeglasses if worn.

3. Garbing & Hand Hygiene — Entry Sequence

Typical order shown for an anteroom with the sink on the dirty side. Your SOP sets the final order; the principle is to don the items from the dirtiest area of the body first (feet, head, face) before hand washing and gowning.

1	**Shoe covers** — don low-lint shoe covers one at a time, stepping each covered foot over the line of demarcation onto the clean side (or use dedicated cleanroom shoes per SOP).

2	**Hair cover** — cover all hair, including ears if possible.
3	**Facial hair cover** — beard/moustache cover if applicable.
4	**Face mask** — cover nose and mouth; adjust nose piece.
5	**Eye protection** — if required by your SOP or for hazardous drugs/cleaning agents.
6	**Hand hygiene** (see Section 4): clean under fingernails, then wash hands and forearms to the elbows for at least 30 seconds; dry completely with low-lint disposable towels.
7	**Gown** — don a low-lint, disposable gown that closes at the neck and fits snugly at the wrists without touching the floor. Do not let it touch the floor or walls.
8	Enter the buffer room.
9	**Alcohol-based hand rub** — apply a waterless, persistent-activity alcohol hand rub to all surfaces of hands and fingers; rub until completely dry.
10	**Sterile gloves** — don sterile, powder-free gloves. Inspect for holes or tears; replace any that are damaged.
11	**Glove disinfection** — apply sterile 70% isopropyl alcohol (IPA) to gloves and let dry before working, and repeat routinely during compounding and whenever gloves touch a non-sterile surface (hair, face, floor, packaging).

Category 3 CSPs: additional garb applies at all times — e.g., sterile low-lint gown (or one laundered/sterilized garment), no exposed skin in the buffer room, sterile sleeves/goggles per SOP, and garb is not reused. **Hazardous drugs:** follow USP <800> (chemo gowns, two pairs of ASTM D6978 gloves, second shoe covers, negative-pressure room).

4. Hand Hygiene Procedure (Detail)

1	Remove debris from under fingernails with a nail cleaner under warm running water.
2	Wash hands and forearms up to the elbows with soap (antimicrobial or plain, per SOP) and water for at least 30 seconds.
3	Rinse thoroughly, keeping hands above elbows so water runs off at the elbow.
4	Dry hands and forearms completely with low-lint disposable towels or wipers. Do not use electric hand dryers.
5	Turn off faucet with a clean towel if not hands-free. Do not touch the sink or other surfaces afterward.
6	Immediately before sterile gloves (in the buffer room), apply alcohol-based hand rub and allow hands to dry completely.

5. Bringing Supplies Into the Cleanroom

1	Remove outer shipping cartons in the unclassified area — cardboard never enters the anteroom or buffer room.
2	In the anteroom, wipe the outer surface of each item with a low-lint wipe wetted with a disinfectant (e.g., 70% IPA); observe the disinfectant's contact time.
3	Pass items into the buffer room via pass-through or the clean side of the anteroom. Bring in only what is needed for the session.
4	Immediately before placing items in the PEC, wipe them with **sterile** 70% IPA and allow to dry. Do not spray directly inside the PEC in a way that dislodges particles onto critical sites.
5	Disinfect critical sites (vial stoppers, ampule necks, injection ports) with sterile 70% IPA and allow to dry before each puncture.

6. Working in the PEC (ISO Class 5)

1	Confirm the PEC is running and certified (certification label within 6 months). If it was turned off, run it per manufacturer instructions and clean/disinfect before use.
2	Clean and disinfect the work surface, then apply sterile 70% IPA to the horizontal work surface immediately before compounding.
3	Work at least 6 inches inside the front opening (LAFW) or per manufacturer for BSCs. Do not block the HEPA filter or air grilles.
4	Maintain first air: never place hands, components or objects between the HEPA filter and a critical site.
5	Arrange items to minimize clutter and motion; keep only items needed for the current CSP inside the hood.
6	Re-disinfect gloves with sterile 70% IPA frequently and after touching anything outside the direct compounding area.
7	Between batches or after spills, clean and disinfect the work surface again; apply sterile 70% IPA before resuming.
8	Label, inspect and verify each CSP per SOP; discard waste in designated containers.

7. Exiting the Cleanroom

1	Remove and discard gloves. Gloves are never reused.
2	Non-HD areas: the gown may be kept for reuse within the same shift only if it stays in the classified area (or SCA perimeter) and is not visibly soiled; hang it per SOP. Otherwise discard.
3	Discard shoe covers, hair cover, beard cover and mask when leaving. New ones are donned on re-entry.
4	Category 3 or hazardous-drug work: discard all garb on exit (per USP <797> / <800>).
5	Re-entry always requires full hand hygiene and new garb as in Section 3.

8. Cleaning, Disinfecting & Sporicidal Schedule

Minimum frequencies (Category 1 and 2). Use low-lint wipers and mops; supplies used inside the PEC must be sterile. Clean first (or use a one-step cleaner-disinfectant), then disinfect, observing the label contact time.

Site	Clean & disinfect	Sporicidal agent
PEC and equipment inside it (all interior surfaces)	Daily on compounding days; when contamination is suspected	Monthly

Site	Clean & disinfect	Sporicidal agent
PEC horizontal work surface	Plus sterile 70% IPA immediately before compounding and during work	Monthly
Removable PEC work tray	Tray surface daily; all surfaces and area underneath monthly	Monthly
Pass-through chambers	Daily on compounding days	Monthly
Work surfaces outside the PEC	Daily on compounding days	Monthly
Floors	Daily on compounding days	Monthly
Walls, doors, door frames	Monthly	Monthly
Ceilings	Monthly	Monthly
Storage shelving and bins	Monthly	Monthly
Equipment outside the PEC (carts, chairs, refrigerators)	Monthly	Monthly

Category 3: sporicidal applied weekly in all classified areas. Clean after any spill, after maintenance or construction work, and if compounding did not take place on a given day, cleaning must be done before compounding resumes.

9. Cleaning Technique

1	Garb fully before cleaning. Cleaning follows the same dirty-to-clean logic as personnel flow.
2	**Order of rooms:** PEC first, then buffer room, then anteroom (cleanest to least clean).
3	**Inside the PEC:** work from top to bottom and from back (HEPA side) to front, using overlapping unidirectional strokes. Clean the ceiling, back wall, IV bar/hooks, side walls, then the work surface. Do not touch the HEPA filter face.
4	**Rooms:** ceilings, then walls, then fixtures and shelving, then floors last. Mop floors from the far wall toward the exit.
5	Use a fresh wiper/side for each surface; never return a used wipe to the disinfectant container.
6	Keep surfaces visibly wet for the full contact time listed on the product label.
7	Apply sterile 70% IPA to the PEC work surface after disinfection/sporicidal to remove residue.
8	Document every cleaning on the facility log (date, time, area, agent, lot, initials).

10. Quality Checks & Personnel Qualification

Requirement	Minimum frequency (Cat 1 & 2)
Garbing competency + gloved fingertip/thumb sampling	Initial: 3 separate times with zero CFU; then every 6 months (Category 3: every 3 months)
Aseptic technique + media-fill test	Initial, then every 6 months (Category 3: every 3 months)
Cleaning & disinfection competency (visual observation)	Initial, then every 12 months
PEC and room certification (airflow, HEPA integrity, particle counts, pressure)	Every 6 months and after relocation or major repair
Viable air sampling	Every 6 months (Category 3: monthly)
Surface sampling	Monthly (Category 3: weekly)

Requirement	Minimum frequency (Cat 1 & 2)
Pressure differential / temperature / humidity	Monitored and documented each compounding day (continuous monitors recommended)

Frequencies summarized from USP <797> (2023); confirm against the current official chapter and your state rules.

Appendix A — Daily Garbing Checklist

Step	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Personal items, jewelry, cosmetics removed							
Shoe covers							
Hair cover							
Beard cover (if applicable)							
Face mask							
Nails cleaned + hands/forearms washed 30 s							
Dried with low-lint towels							
Gown donned and closed							
Alcohol hand rub, hands fully dry							
Sterile gloves donned and inspected							
Gloves disinfected with sterile 70% IPA							

Employee name	Signature	Week of

Appendix B — Cleaning Log

Date	Time	Area / site	Cleaner / disinfectant (lot)	Sporicidal ?	IPA?	Initials